BUILDING AND CONSTRUCTION

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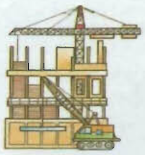


THE SIMPLEST BUILDING is a permanent structure with a roof and four walls. Buildings come in a huge variety of shapes, sizes, and appearances – from skyscrapers and factories to schools, hospitals, houses, and garden sheds. Despite these differences, all buildings have the same basic purpose – to provide a sheltered area in which people can live, work, or store belongings. The engineers, surveyors, and construction workers who plan and build these structures also work on other projects, such as roads, bridges, dams, and tunnels.

Ancient tower-house, Sana, Yemen

Early building

Since the beginning of history, people have built shelters to protect themselves from the weather, wild animals, and their enemies. The first buildings were simple, single-storey structures made of materials such as wood, stone, and dried grass and mud. The first large-scale stone constructions were temples for the worship of gods and goddesses, and palaces in which powerful leaders lived. About 6,000 years ago, people discovered how to bake clay bricks. In time, engineers developed new building methods that enabled them to build higher and lighter structures.

Walls are made from mud and bricks dried in the Sun's heat.

Anatomy of a building

Most buildings have certain features in common, such as walls, a roof, and floors. A large modern building, such as this airport terminal, also has a strong internal frame. Underneath this are the solid foundations on which the whole structure rests. The building is equipped with services, such as electricity and water supplies, as well as escalators, stairs, or elevators to give access to different storeys, and fire escapes that enable people to leave the building rapidly in the event of an emergency.

Roof

A roof is a protective covering over a building. Roofing materials include thatch, clay tiles, slate, glass, and steel. Roofs in wet climates are shaped to make rainwater run off; in cold countries, they slope steeply to stop snow from building up; and in dry climates, they are often flat. Sloping roofs are held up by supports called roof trusses.

Overhead cutaway of roof

Kansai Airport, Japan

Foundations

A building's foundations spread its huge load evenly into the ground, stopping the building from sinking under its own weight. Pile foundations are columns that rest on hard rock; raft foundations are concrete platforms that rest on soft rock. The foundations form the base on which the building's frame is constructed.

Internal frame

The "skeleton" of a large building is its internal frame, which supports the roof, the walls, and the floors. Frames can be made of wood, steel, or reinforced-concrete columns and beams joined together.

Walls and floors

In a house, the walls – which may be made of wood, stone, or brick – are strong enough to hold up the floors, ceilings, and roof trusses. In a larger structure, however, the frame supports the building's weight, and the walls simply hang from the frame. The floors in a large modern building are reinforced-concrete slabs.

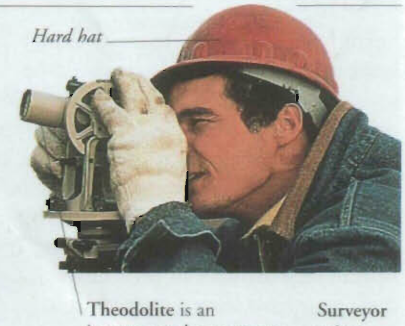
Structural engineers

Long before the construction of a building is underway, structural engineers begin working on the design of the building with an architect. They calculate how strong the building's structure needs to be and draw up detailed plans, usually on a computer. When the building work commences, they make sure that everything happens safely, on time, and within the financial budget.

Structural engineer on a building site

Surveyors

Accuracy is extremely important in construction work if the completed building is to have vertical sides and level walls, and be structurally safe. Even small errors in the design or assembly can result in parts not fitting together properly. People called surveyors check the building at every stage of its construction, using special instruments, such as theodolites and spirit levels, to take accurate measurements.



Theodolite is an instrument that measures angles to find distances, lengths, and heights.

Surveyor using theodolite